Statistics/Data analysis

```
      name:  The_value_of_a_helping_hand_sup
       log:  C:\Users\hansg\Documents\working\SHARE data\weighting\output_sup_5.smcl
  log type:  smcl
 opened on:   5 Apr 2026, 14:31:11
C:\Users\hansg\Documents\working\SHARE data\weighting
(1 observation deleted)
(10,395 real changes made)
(6,152 real changes made)
(70,175 real changes made)
(8,277 real changes made)
(23,812 real changes made)
(1,763 real changes made)
(1,201 real changes made)
(1,763 real changes made)
```

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test Adj chi2(2)	Joint test Prob>chi2
age	109,546	0.0000	0.0000	.	.

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test Adj chi2(2)	Joint test Prob>chi2
thinc	109,546	0.0000	0.0000	.	.

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test Adj chi2(2)	Joint test Prob>chi2
yedu	109,546	0.0000	0.0000	106.08	0.0000

(11,207 real changes made)

Summary for variables: id
Group variable: htype (Household type)

htype	N
1 S	37717
1 CNRP	8553
1 C2R	62737
1 CNRP + 1 S	17
Multi S	372
1 C2R + 1 S	117
1 C2R + 1 CNRP	2
Multi C2R	29
Multi S + 1 C2R	2
Total	109546

Household type	Freq.	Percent	Cum.
1	37,717	34.43	34.43
2	8,553	7.81	42.24
3	62,737	57.27	99.51
4	17	0.02	99.52
5	372	0.34	99.86
6	117	0.11	99.97
7	2	0.00	99.97
8	29	0.03	100.00
10	2	0.00	100.00
Total	109,546	100.00	

(37,717 real changes made)
(62,737 real changes made)

```
note: 2.naly omitted because of collinearity.
note: 2.rhfo#0.ghto identifies no observations in the sample.
note: 2.rhfo#1.ghto identifies no observations in the sample.
note: 2.rhfo#2.ghto omitted because of collinearity.
Iteration 0:  Log likelihood = -155917.03
Iteration 1:  Log likelihood = -129392.98
Iteration 2:  Log likelihood = -127789.78
Iteration 3:  Log likelihood = -127780.08
Iteration 4:  Log likelihood = -127780.08

Ordered logistic regression                         Number of obs =  109,546
                                                    LR chi2(37)   = 56273.89
                                                    Prob > chi2   =   0.0000
Log likelihood = -127780.08                         Pseudo R2     =   0.1805
```

sphus	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]	
age	.0840308	.0136638	6.15	0.000	.0572502	.1108113
c.age#c.age	-.000381	.0000904	-4.21	0.000	-.0005583	-.0002038
yedu	-.0053371	.0049421	-1.08	0.280	-.0150234	.0043493
c.yedu#c.yedu	-.0015312	.0002162	-7.08	0.000	-.001955	-.0011075
thinc	-.1984827	.0138307	-14.35	0.000	-.2255905	-.171375
c.thinc#c.thinc	.0276869	.006187	4.48	0.000	.0155606	.0398132
gender						
Female	-.0227	.0126334	-1.80	0.072	-.047461	.002061
1.covid	-.1164656	.0196685	-5.92	0.000	-.1550152	-.077916
household						
Couple, both responding	.1774136	.0152921	11.60	0.000	.1474416	.2073855
Multiple, at least 1 responding	.2057342	.0242593	8.48	0.000	.1581869	.2532815
cjs						
Not applicable	.5937323	.0587032	10.11	0.000	.4786761	.7087885
Retired	.3234076	.0295527	10.94	0.000	.2654854	.3813298
Unemployed	.2371544	.0878396	2.70	0.007	.064992	.4093168
Permanently sick or disabled	1.445825	.0628292	23.01	0.000	1.322682	1.568969
Homemaker	.3824268	.0362272	10.56	0.000	.3114228	.4534307
Other	.233202	.062453	3.73	0.000	.1107964	.3556076
gali						
Limited	1.991348	.0139059	143.20	0.000	1.964093	2.018603
lifesat	-.3087225	.0189354	-16.30	0.000	-.3458352	-.2716098
c.lifesat#c.lifesat	-.0004697	.0013185	-0.36	0.722	-.003054	

```
>     .0021146
                                  |
                            naly  |
                     At least 1   |  -.3638971    .019021   -19.13   0.000    -.4011776
>    -.3266165
                  Not applicable  |          0  (omitted)
                                  |
                            rhfo  |
                     At least 1   |   .4026307   .0184658    21.80   0.000     .3664384
>     .4388231
                  Not applicable  |   -.179064   .0325387    -5.50   0.000    -.2428387
>    -.1152894
                                  |
                            ghto  |
                     At least 1   |  -.2336388   .0181786   -12.85   0.000    -.2692681
>    -.1980094
                  Not applicable  |   .0235166   .0197795     1.19   0.234    -.0152506
>     .0622838
                                  |
                      rhfo#ghto   |
          At least 1#At least 1   |  -.1945523   .0334787    -5.81   0.000    -.2601693
>    -.1289353
      At least 1#Not applicable   |  -.0430327     .04618    -0.93   0.351    -.1335439
>     .0474785
            Not applicable#None   |          0  (empty)
      Not applicable#At least 1   |          0  (empty)
  Not applicable#Not applicable   |          0  (omitted)
                                  |
                         country  |
                         Austria  |   .1860885   .0296468     6.28   0.000     .1279817
>     .2441952
                         Germany  |   .6518496   .0365225    17.85   0.000     .5802669
>     .7234324
                     Netherlands  |   .0732863   .0359976     2.04   0.042     .0027322
>     .1438403
                           Spain  |   .7725628   .0337869    22.87   0.000     .7063416
>     .8387839
                           Italy  |   .5720736    .032754    17.47   0.000     .5078769
>     .6362703
                          France  |   .4944956   .0295805    16.72   0.000     .4365188
>     .5524723
                         Denmark  |  -.2629128   .0348731    -7.54   0.000    -.3312629
>    -.1945627
                     Switzerland  |   .0660177   .0320577     2.06   0.039     .0031857
>     .1288496
                         Belgium  |    .111085   .0296281     3.75   0.000     .0530149
>     .1691551
                  Czech Republic  |   .4195933    .031893    13.16   0.000     .3570842
>     .4821024
                        Slovenia  |   .5819763   .0348995    16.68   0.000     .5135745
>     .6503781
------------------------------------+----------------------------------------------------------------
  ------------
                           /cut1  |   -.733932   .5122886                      -1.737999
>     .2701352
                           /cut2  |   1.046156   .5123795                       .0419104
>     2.050401
                           /cut3  |   3.534302   .5126266                       2.529572
>     4.539031
                           /cut4  |   5.826479   .5126037                       4.821794
>     6.831164
------------------------------------+----------------------------------------------------------------
  ------------
```

Tests of proportional odds assumption

```
Number of obs            = 109,546
Number of predictors     =      37
Number of outcome levels =       5
Degrees of freedom       =     111
```

Test	chi2	P>chi2
Brant	**4063.76**	**0.000**
Likelihood-ratio	**3994.02**	**0.000**
Score	**4099.53**	**0.000**
Wald	**4124.33**	**0.000**
Wolfe-Gould	**4034.24**	**0.000**

(file **TestAssum_5.doc** not found)
Click to Open File: TestAssum_5.doc
note: **2.naly** omitted because of collinearity.
note: **2.rhfo#0.ghto** identifies no observations in the sample.
note: **2.rhfo#1.ghto** identifies no observations in the sample.
note: **2.rhfo#2.ghto** omitted because of collinearity.

Generalized Ordered Logit Estimates Number of obs = **109,528**
Wald chi2(1) = .
Prob > chi2 = .
Log pseudolikelihood = **-122318.19** Pseudo R2 = **0.1985**

(Std. err. adjusted for **6** cluste
> rs in **Qyear**)

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% con
Excellent					
age	**.1049038**	**.0842558**	**1.25**	**0.213**	**-.0602345**
> **.2700422**					
c.age#c.age	**-.0004962**	**.0005966**	**-0.83**	**0.406**	**-.0016655**
> **.0006731**					
yedu	**.0320913**	**.0169606**	**1.89**	**0.058**	**-.0011508**
> **.0653335**					
c.yedu#c.yedu	**-.0019413**	**.0008**	**-2.43**	**0.015**	**-.0035093**
> **-.0003734**					
thinc	**-.1538899**	**.0503431**	**-3.06**	**0.002**	**-.2525605**
> **-.0552193**					
c.thinc#c.thinc	**.0134405**	**.03322**	**0.40**	**0.686**	**-.0516694**
> **.0785504**					
gender					
Female	**.1453991**	**.06064**	**2.40**	**0.016**	**.026547**
> **.2642513**					
covid					
1	**-.0068584**	**.0124182**	**-0.55**	**0.581**	**-.0311976**
> **.0174809**					
household					
Couple, both responding	**.1847433**	**.0582257**	**3.17**	**0.002**	**.070623**
> **.2988636**					
Multiple, at least 1 responding	**.1989461**	**.0751503**	**2.65**	**0.008**	**.0516542**
> **.346238**					
cjs					
Not applicable	**.3137035**	**.5412106**	**0.58**	**0.562**	**-.7470498**
> **1.374457**					
Retired	**.2870771**	**.1154882**	**2.49**	**0.013**	**.0607245**
> **.5134297**					
Unemployed	**.0909656**	**.2685006**	**0.34**	**0.735**	**-.4352859**
> **.617217**					
Permanently sick or disabled	**.5691217**	**.1340084**	**4.25**	**0.000**	**.3064701**
> **.8317732**					
Homemaker	**.3765284**	**.1968216**	**1.91**	**0.056**	**-.0092347**
> **.7622916**					

```
                          Other |  -.2504163   .3050821    -0.82   0.412    -.8483662
>     .3475337
                                |
                           gali |
                        Limited |   2.027874   .0432075    46.93   0.000     1.943189
>     2.112559
                                |
                        lifesat |   .5006842   .0867741     5.77   0.000     .3306102
>     .6707583
                                |
             c.lifesat#c.lifesat |  -.0618902   .0056412   -10.97   0.000    -.0729469
>    -.0508336
                                |
                           naly |
                      At least 1 |  -.1664662   .0198042    -8.41   0.000    -.2052816
>    -.1276508
                                |
                           rhfo |
                      At least 1 |   .2031786   .0630515     3.22   0.001     .0795999
>     .3267573
                  Not applicable |  -.2736854   .0440193    -6.22   0.000    -.3599616
>    -.1874092
                                |
                           ghto |
                      At least 1 |  -.1996347   .0930743    -2.14   0.032    -.3820569
>    -.0172124
                  Not applicable |  -.0491722   .0301916    -1.63   0.103    -.1083467
>     .0100022
                                |
                       rhfo#ghto |
          At least 1#At least 1 |  -.2035065   .0802705    -2.54   0.011    -.3608338
>    -.0461791
       At least 1#Not applicable |  -.1849287   .0554822    -3.33   0.001    -.2936719
>    -.0761856
                                |
                        country |
                        Austria |   .7265466   .0864409     8.41   0.000     .5571256
>     .8959676
                        Germany |    1.35485   .1170965    11.57   0.000     1.125345
>     1.584355
                    Netherlands |  -.1258238   .0960822    -1.31   0.190    -.3141415
>     .0624939
                          Spain |   1.520409   .0664546    22.88   0.000      1.39016
>     1.650658
                          Italy |     .63835   .0865324     7.38   0.000     .4687496
>     .8079505
                         France |    .668267   .0410348    16.29   0.000     .5878403
>     .7486938
                        Denmark |   .0370801    .061261     0.61   0.545    -.0829892
>     .1571494
                    Switzerland |   .8583194    .120059     7.15   0.000      .623008
>     1.093631
                        Belgium |   .6555846   .0603675    10.86   0.000     .5372665
>     .7739027
                  Czech Republic |   1.359282   .1842497     7.38   0.000     .9981598
>     1.720405
                       Slovenia |   1.022731   .1966819     5.20   0.000     .6372414
>      1.40822
                                |
                          _cons |  -3.442279   3.184921    -1.08   0.280    -9.684609
>     2.800051
------------------------------+----------------------------------------------------------
  ____________
Very_good                       |
                            age |   .0407537    .032494     1.25   0.210    -.0229334
>     .1044409
                                |
                    c.age#c.age |  -.0000452   .0002118    -0.21   0.831    -.0004603
>     .0003699
                                |
                           yedu |   .0101751    .017727     0.57   0.566    -.0245691
>     .0449193
```

```
                          |
            c.yedu#c.yedu |  -.0025317   .0007603    -3.33   0.001    -.0040219
>    -.0010414
                          |
                    thinc |  -.1879521   .0559984    -3.36   0.001    -.2977069
>    -.0781973
                          |
        c.thinc#c.thinc   |    .002208   .0286699     0.08   0.939     -.053984
>        .0584
                          |
                   gender |
                   Female |   .0629036   .0235653     2.67   0.008     .0167164
>     .1090908
                          |
                    covid |
                        1 |  -.0583106    .028783    -2.03   0.043    -.1147243
>    -.0018968
                          |
                household |
  Couple, both responding |   .2271396   .0264545     8.59   0.000     .1752896
>     .2789895
Multiple, at least 1 responding |   .2720259   .0272881     9.97   0.000     .2185423
>     .3255095
                          |
                      cjs |
            Not applicable |   .2443961   .3823192     0.64   0.523    -.5049358
>     .9937281
                   Retired |   .3377246   .0623222     5.42   0.000     .2155753
>      .459874
                Unemployed |   .2153731   .1285993     1.67   0.094    -.0366768
>      .467423
 Permanently sick or disabled |   .6063775   .1566321     3.87   0.000     .2993843
>     .9133707
                 Homemaker |   .3038846   .0468557     6.49   0.000     .2120491
>     .3957202
                     Other |  -.1763158   .3456531    -0.51   0.610    -.8537834
>     .5011517
                          |
                      gali |
                   Limited |   1.830286   .0313836    58.32   0.000     1.768775
>     1.891797
                          |
                   lifesat |    .064052   .0938979     0.68   0.495    -.1199845
>     .2480884
                          |
      c.lifesat#c.lifesat |  -.0278149   .0060571    -4.59   0.000    -.0396866
>    -.0159432
                          |
                      naly |
                 At least 1 |   .0214308   .0438782     0.49   0.625     -.064569
>     .1074306
                          |
                      rhfo |
                 At least 1 |   .1757839   .0404798     4.34   0.000      .096445
>     .2551228
            Not applicable |  -.1615497   .0658534    -2.45   0.014      -.29062
>    -.0324794
                          |
                      ghto |
                 At least 1 |  -.1424987   .0334354    -4.26   0.000    -.2080308
>    -.0769665
            Not applicable |  -.0119509   .0392789    -0.30   0.761    -.0889361
>     .0650343
                          |
                 rhfo#ghto |
   At least 1#At least 1 |  -.1014996   .1464483    -0.69   0.488    -.3885331
>     .1855338
  At least 1#Not applicable |  -.1049326   .0530384    -1.98   0.048    -.2088859
>    -.0009793
                          |
                   country |
                   Austria |   .2481899   .0405618     6.12   0.000     .1686902
```

```
>     .3276896
                    Germany |   1.101716   .0682749    16.14   0.000     .9678992
>     1.235532
                Netherlands |   .3831335   .0546152     7.02   0.000     .2760896
>     .4901774
                      Spain |   1.054148   .0479521    21.98   0.000     .9601639
>     1.148133
                      Italy |   .8962374   .0456567    19.63   0.000     .8067519
>      .985723
                     France |   .8666234   .0317121    27.33   0.000     .8044687
>     .9287781
                    Denmark |  -.3693694    .057772    -6.39   0.000    -.4826006
>    -.2561383
                Switzerland |   .3929763   .0480379     8.18   0.000     .2988237
>     .4871288
                    Belgium |   .4994658   .0566509     8.82   0.000      .388432
>     .6104996
            Czech Republic  |   .9402304    .067648    13.90   0.000     .8076427
>     1.072818
                   Slovenia |   .9973119   .1134343     8.79   0.000     .7749847
>     1.219639
                            |
                      _cons |  -1.499096   1.145418    -1.31   0.191    -3.744074
>     .7458816
------------------------------+----------------------------------------------------------------
  ____________
Good                        |
                        age |   .1407329   .0286931     4.90   0.000     .0844954
>     .1969704
                            |
                 c.age#c.age |  -.0007878   .0001918    -4.11   0.000    -.0011636
>    -.0004119
                            |
                       yedu |  -.0291724   .0068721    -4.25   0.000    -.0426414
>    -.0157034
                            |
               c.yedu#c.yedu |  -.0006772   .0003767    -1.80   0.072    -.0014155
>      .000061
                            |
                      thinc |  -.2049144    .035367    -5.79   0.000    -.2742324
>    -.1355964
                            |
            c.thinc#c.thinc |   .0339365   .0174681     1.94   0.052    -.0003003
>     .0681734
                            |
                     gender |
                     Female |  -.0144844     .01945    -0.74   0.456    -.0526056
>     .0236368
                            |
                      covid |
                          1 |  -.1095895   .0239281    -4.58   0.000    -.1564877
>    -.0626914
                            |
                  household |
        Couple, both responding |   .1424639   .0446205     3.19   0.001     .0550094
>     .2299185
Multiple, at least 1 responding |   .2587334    .038887     6.65   0.000     .1825162
>     .3349505
                            |
                        cjs |
            Not applicable  |   .4378829   .2540835     1.72   0.085    -.0601116
>     .9358775
                    Retired |   .2718471   .1088822     2.50   0.013     .0584418
>     .4852523
                 Unemployed |   .1447865    .107411     1.35   0.178    -.0657352
>     .3553083
   Permanently sick or disabled |   1.004384   .0488585    20.56   0.000     .9086227
>     1.100145
                  Homemaker |   .2939988   .1901028     1.55   0.122    -.0785959
>     .6665934
                      Other |   .0128281   .2626526     0.05   0.961    -.5019615
>     .5276178
```

```
                                 |
                        gali     |
                     Limited     |   2.108183   .0166402   126.69   0.000     2.075569
>     2.140797
                                 |
                     lifesat     |   -.275062   .0584971    -4.70   0.000    -.3897143
>    -.1604097
                                 |
           c.lifesat#c.lifesat   |  -.0020662   .0039003    -0.53   0.596    -.0097106
>     .0055782
                                 |
                        naly     |
                   At least 1    |  -.2148956   .0513777    -4.18   0.000    -.3155941
>    -.1141972
                                 |
                        rhfo     |
                   At least 1    |   .4021789    .061449     6.54   0.000     .2817411
>     .5226167
              Not applicable     |  -.1277717   .0493217    -2.59   0.010    -.2244404
>    -.0311031
                                 |
                        ghto     |
                   At least 1    |  -.2403215   .0526048    -4.57   0.000     -.343425
>    -.1372181
              Not applicable     |  -.0068664   .0275936    -0.25   0.803     -.060949
>     .0472161
                                 |
                    rhfo#ghto    |
       At least 1#At least 1     |  -.1301424   .1432367    -0.91   0.364    -.4108811
>     .1505963
     At least 1#Not applicable   |  -.0410424   .0677692    -0.61   0.545    -.1738676
>     .0917828
                                 |
                     country     |
                     Austria     |  -.0371929   .0970847    -0.38   0.702    -.2274754
>     .1530896
                     Germany     |   .296959   .0680943     4.36   0.000     .1634965
>     .4304214
                 Netherlands     |   -.117236   .0398816    -2.94   0.003    -.1954026
>    -.0390695
                       Spain     |   .5721343   .0699435     8.18   0.000     .4350476
>      .709221
                       Italy     |   .5236342    .109236     4.79   0.000     .3095356
>     .7377328
                      France     |   .1135564   .0484309     2.34   0.019     .0186335
>     .2084793
                     Denmark     |  -.0700092   .0888035    -0.79   0.430    -.2440608
>     .1040425
                 Switzerland     |  -.3727606   .0848896    -4.39   0.000    -.5391412
>    -.2063801
                     Belgium     |  -.2754552   .0363477    -7.58   0.000    -.3466954
>    -.2042149
              Czech Republic     |  -.1399726   .1252739    -1.12   0.264    -.3855049
>     .1055597
                    Slovenia     |   .1804686   .0856404     2.11   0.035     .0126165
>     .3483208
                                 |
                       _cons     |  -5.378249   1.014638    -5.30   0.000    -7.366902
>    -3.389596
-----------------------------------+----------------------------------------------------------------
  ____________
Fair                             |
                         age     |   .1712421   .0425605     4.02   0.000      .087825
>     .2546592
                                 |
                   c.age#c.age   |  -.0010067   .0002816    -3.57   0.000    -.0015586
>    -.0004547
                                 |
                        yedu     |  -.0128055   .0095198    -1.35   0.179    -.0314639
>     .0058528
                                 |
                  c.yedu#c.yedu  |  -.0012419   .0006437    -1.93   0.054    -.0025034
```

```
>     .0000197
                                  |
                           thinc  |  -.1681867   .0600294    -2.80   0.005    -.2858421
>    -.0505313
                                  |
                c.thinc#c.thinc   |   .0469303   .0352898     1.33   0.184    -.0222365
>     .1160971
                                  |
                          gender  |
                          Female  |   -.081211   .0531965    -1.53   0.127    -.1854742
>     .0230523
                                  |
                           covid  |
                               1  |  -.1085464   .0462885    -2.34   0.019    -.1992702
>    -.0178227
                                  |
                       household  |
        Couple, both responding   |   .1267859    .055582     2.28   0.023     .0178472
>     .2357245
Multiple, at least 1 responding   |   .0975863    .057154     1.71   0.088    -.0144334
>      .209606
                                  |
                             cjs  |
                Not applicable    |   .4980276   .1819926     2.74   0.006     .1413288
>     .8547265
                        Retired   |   .2876118   .0847471     3.39   0.001     .1215107
>      .453713
                      Unemployed  |  -.7204413     .14874    -4.84   0.000    -1.011966
>    -.4289164
   Permanently sick or disabled   |   1.428278   .0962504    14.84   0.000      1.23963
>     1.616925
                       Homemaker  |   .2401469   .0930793     2.58   0.010     .0577148
>      .422579
                           Other  |   .1133735   .1484529     0.76   0.445    -.1775887
>     .4043358
                                  |
                            gali  |
                         Limited  |   2.978319   .0742228    40.13   0.000     2.832845
>     3.123793
                                  |
                          lifesat  |  -.5310374    .052949   -10.03   0.000    -.6348155
>    -.4272592
                                  |
             c.lifesat#c.lifesat  |   .0176566   .0033813     5.22   0.000     .0110294
>     .0242838
                                  |
                             naly  |
                       At least 1  |  -.5280083    .092357    -5.72   0.000    -.7090247
>     -.346992
                                  |
                             rhfo  |
                       At least 1  |   .5810003   .0309895    18.75   0.000     .5202619
>     .6417387
                Not applicable    |  -.1527783   .0951531    -1.61   0.108     -.339275
>     .0337184
                                  |
                             ghto  |
                       At least 1  |  -.5506786   .0785023    -7.01   0.000    -.7045402
>     -.396817
                Not applicable    |   .1486777   .0288475     5.15   0.000     .0921376
>     .2052178
                                  |
                       rhfo#ghto  |
          At least 1#At least 1   |   -.007915   .1600186    -0.05   0.961    -.3215457
>     .3057157
      At least 1#Not applicable   |   .0387761   .0330458     1.17   0.241    -.0259925
>     .1035446
                                  |
                          country |
                          Austria |  -.1264017   .2000582    -0.63   0.528    -.5185086
>     .2657052
                          Germany |   .0426663   .1628714     0.26   0.793    -.2765557
```

```
>     .3618883
                 Netherlands |  -.4489427   .1506674    -2.98   0.003    -.7442453
>    -.1536401
                       Spain |   .480052   .1538039     3.12   0.002     .1786019
>     .7815022
                       Italy |  -.085668   .1354414    -0.63   0.527    -.3511283
>     .1797923
                      France |   .2264456   .1569828     1.44   0.149     -.081235
>     .5341263
                     Denmark |  -.000336   .1908093    -0.00   0.999    -.3743153
>     .3736433
                 Switzerland |  -.4771648   .1814002    -2.63   0.009    -.8327027
>    -.1216269
                     Belgium |  -.4962474   .0886511    -5.60   0.000    -.6700005
>    -.3224944
             Czech Republic |  -.0296629    .194436    -0.15   0.879    -.4107503
>     .3514246
                    Slovenia |   .3426955   .1246779     2.75   0.006     .0983314
>     .5870597
                             |
                       _cons |  -8.550824   1.674413    -5.11   0.000    -11.83261
>    -5.269036
------------------------------+----------------------------------------------------
  ____________

WARNING! 7 in-sample cases have an outcome with a predicted probability that is
less than 0. See the gologit2 help section on Warning Messages for more information.
results_imp1_go_5.xls
dir : seeout
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -2.853e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.794e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.298e+08  (not concave)
Iteration 3:  Log pseudolikelihood = -1.030e+08  (not concave)
Iteration 4:  Log pseudolikelihood =  -89777458  (not concave)
Iteration 5:  Log pseudolikelihood =  -85679267  (not concave)
Iteration 6:  Log pseudolikelihood =  -82800728  (not concave)
Iteration 7:  Log pseudolikelihood =  -81428306  (not concave)
Iteration 8:  Log pseudolikelihood =  -80219837  (not concave)
Iteration 9:  Log pseudolikelihood =  -79899022  (not concave)
Iteration 10: Log pseudolikelihood =  -79604702  (not concave)
Iteration 11: Log pseudolikelihood =  -79123162  (not concave)
Iteration 12: Log pseudolikelihood =  -78766254  (not concave)
Iteration 13: Log pseudolikelihood =  -78606452  (not concave)
Iteration 14: Log pseudolikelihood =  -78492643  (not concave)
Iteration 15: Log pseudolikelihood =  -78399081
Iteration 16: Log pseudolikelihood =  -78137708  (not concave)
Iteration 17: Log pseudolikelihood =  -78085449
Iteration 18: Log pseudolikelihood =  -77796400
Iteration 19: Log pseudolikelihood =  -77755602
Iteration 20: Log pseudolikelihood =  -77754238
Iteration 21: Log pseudolikelihood =  -77754224
Iteration 22: Log pseudolikelihood =  -77754224

Stereotype logistic regression                      Number of obs =  29,989
                                                    Wald chi2(35) = 2254.40
Log pseudolikelihood = -77754224                    Prob > chi2   =  0.0000

 ( 1)  [phi1_1]_cons = 1
------------------------------+----------------------------------------------------
  ____________
                              |               Robust
                        sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]
------------------------------+----------------------------------------------------
  ____________
                          age |   .2424613   .1292746     1.88   0.061    -.0109122
>     .4958348
                              |
                   c.age#c.age |  -.001183   .0008863    -1.33   0.182      -.00292
```

```
>     .0005541
                                  |
                            yedu  |   .0205388    .039005     0.53   0.598    -.0559097
>     .0969873
                                  |
                  c.yedu#c.yedu   |  -.0066306   .0018065    -3.67   0.000    -.0101713
>    -.0030899
                                  |
                           thinc  |   -.526568   .1121265    -4.70   0.000    -.7463319
>     -.306804
                                  |
                 c.thinc#c.thinc  |   .1038257     .05155     2.01   0.044     .0027896
>     .2048617
                                  |
                          gender  |
                          Female  |    .050958   .1076236     0.47   0.636    -.1599804
>     .2618964
                          0.covid |          0  (omitted)
                                  |
                       household  |
        Couple, both responding   |   .3698112   .1418562     2.61   0.009     .0917781
>     .6478444
Multiple, at least 1 responding   |   .6851191   .1930501     3.55   0.000     .3067478
>      1.06349
                                  |
                             cjs  |
                 Not applicable   |   .1139006   .8579402     0.13   0.894    -1.567631
>     1.795433
                         Retired  |   .6424923    .210853     3.05   0.002     .2292281
>     1.055757
                      Unemployed  |   -.200588    .414702    -0.48   0.629    -1.013389
>     .6122129
   Permanently sick or disabled   |    3.03899   .4454308     6.82   0.000     2.165962
>     3.912019
                       Homemaker  |   .4490062   .2442432     1.84   0.066    -.0297017
>      .927714
                           Other  |  -1.733115   .4867019    -3.56   0.000    -2.687033
>    -.7791968
                                  |
                            gali  |
                         Limited  |   5.385471    .155078    34.73   0.000     5.081524
>     5.689418
                          lifesat |  -.4630053   .1859455    -2.49   0.013    -.8274517
>    -.0985589
                                  |
             c.lifesat#c.lifesat  |  -.0218502   .0130937    -1.67   0.095    -.0475133
>      .003813
                                  |
                            naly  |
                       At least 1 |  -.9210391    .140359    -6.56   0.000    -1.196138
>    -.6459404
                 Not applicable   |          0  (omitted)
                                  |
                            rhfo  |
                       At least 1 |   1.162178   .1773219     6.55   0.000     .8146335
>     1.509722
                                  |
                            ghto  |
                       At least 1 |  -.4609722   .1539645    -2.99   0.003    -.7627371
>    -.1592073
                 Not applicable   |  -.0613369   .1240859    -0.49   0.621    -.3045408
>      .181867
                                  |
                       rhfo#ghto  |
         At least 1#At least 1    |  -1.018538   .3353176    -3.04   0.002    -1.675748
>    -.3613278
      At least 1#Not applicable   |  -.0391844   .2890335    -0.14   0.892    -.6056797
>     .5273109
                                  |
                         country  |
                         Austria  |  -.2555815   .1939471    -1.32   0.188    -.6357109
>     .1245479
```

```
                    Germany |   1.064482   .2220026     4.79   0.000     .6293646
>     1.499599
                Netherlands |  -.1802723   .2082804    -0.87   0.387    -.5884943
>     .2279497
                      Spain |   2.030881   .2246431     9.04   0.000     1.590588
>     2.471173
                      Italy |   1.072604   .2142774     5.01   0.000     .6526277
>     1.492579
                     France |     .92343   .1935688     4.77   0.000     .5440421
>     1.302818
                    Denmark |  -1.293367   .2412973    -5.36   0.000    -1.766301
>    -.8204327
                Switzerland |  -.6188578   .2189588    -2.83   0.005    -1.048009
>    -.1897065
                    Belgium |  -.0465844   .1940261    -0.24   0.810    -.4268686
>     .3336998
            Czech Republic  |   .7468065   .2317308     3.22   0.001     .2926226
>     1.200991
                   Slovenia |    1.41276   .2754915     5.13   0.000     .8728061
>     1.952713
------------------------------+----------------------------------------------------
  ___________
                    /phi1_1 |          1  (constrained)
                    /phi1_2 |   .8526347    .017928    47.56   0.000     .8174966
>     .8877729
                    /phi1_3 |   .6267069    .013323    47.04   0.000     .6005943
>     .6528195
                    /phi1_4 |   .3459663   .0119271    29.01   0.000     .3225896
>      .369343
                    /phi1_5 |          0  (base outcome)
------------------------------+----------------------------------------------------
  ___________
                    /theta1 |   9.003931    4.72472     1.91   0.057    -.2563495
>     18.26421
                    /theta2 |   9.059582   4.032846     2.25   0.025     1.155349
>     16.96381
                    /theta3 |   8.478444   2.979381     2.85   0.004     2.638963
>     14.31792
                    /theta4 |    5.45242   1.660882     3.28   0.001     2.197151
>     8.707689
                    /theta5 |          0  (base outcome)
------------------------------+----------------------------------------------------
  ___________
(sphus=Poor is the base outcome)
results_imp1_ste_11_5.xls
dir : seeout

Predictive margins                                      Number of obs = 29,989
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------+----------------------------------------------------
  ________
                              |            Delta-method
                              |     Margin   std. err.      z    P>|z|     [95% conf. i
> nterval]
------------------------------+----------------------------------------------------
  ________
          _predict#rhfo#ghto  |
                 1#None#None  |   .0527457   .0021159    24.93   0.000     .0485987
> .0568927
            1#None#At least 1 |   .0606689   .0030113    20.15   0.000     .0547669
> .0665708
        1#None#Not applicable |   .0537547   .0024014    22.39   0.000     .0490481
> .0584613
            1#At least 1#None |   .0361193   .0025393    14.22   0.000     .0311425
> .0410962
```

1#At least 1#At least 1	.0581153	.0049971	11.63	0.000	.0483211
> .0679095					
1#At least 1#Not applicable	.037378	.003223	11.60	0.000	.0310611
> .0436949					
2#None#None	.1323751	.0034056	38.87	0.000	.1257002
> .13905					
2#None#At least 1	.1446742	.0046997	30.78	0.000	.135463
> .1538855					
2#None#Not applicable	.1339905	.0038266	35.02	0.000	.1264906
> .1414905					
2#At least 1#None	.10327	.0046097	22.40	0.000	.0942351
> .1123049					
2#At least 1#At least 1	.1408041	.0075315	18.70	0.000	.1260427
> .1555655					
2#At least 1#Not applicable	.1056615	.0058922	17.93	0.000	.0941131
> .11721					
3#None#None	.388563	.0048832	79.57	0.000	.3789921
> .3981339					
3#None#At least 1	.3978178	.0054346	73.20	0.000	.3871663
> .4084693					
3#None#Not applicable	.3898868	.0051461	75.76	0.000	.3798006
> .3999729					
3#At least 1#None	.3585708	.0061475	58.33	0.000	.346522
> .3706196					
3#At least 1#At least 1	.3951077	.00683	57.85	0.000	.3817211
> .4084944					
3#At least 1#Not applicable	.3615077	.0077012	46.94	0.000	.3464135
> .3766018					
4#None#None	.3096914	.0046457	66.66	0.000	.3005859
> .3187969					
4#None#At least 1	.2960831	.005682	52.11	0.000	.2849466
> .3072196					
4#None#Not applicable	.3079438	.005052	60.96	0.000	.2980421
> .3178454					
4#At least 1#None	.3385879	.006279	53.92	0.000	.3262814
> .3508944					
4#At least 1#At least 1	.3004377	.008814	34.09	0.000	.2831625
> .3177128					
4#At least 1#Not applicable	.3364299	.006915	48.65	0.000	.3228768
> .349983					
5#None#None	.1166249	.0036729	31.75	0.000	.1094262
> .1238236					
5#None#At least 1	.100756	.005051	19.95	0.000	.0908562
> .1106558					
5#None#Not applicable	.1144242	.0043358	26.39	0.000	.1059263
> .1229222					
5#At least 1#None	.163452	.007115	22.97	0.000	.1495068
> .1773972					
5#At least 1#At least 1	.1055352	.0089937	11.73	0.000	.0879078
> .1231626					
5#At least 1#Not applicable	.1590229	.0101087	15.73	0.000	.1392102
> .1788356					

slog11_5.xls
dir : seeout
note: **0.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -2.299e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.929e+08  (not concave)
Iteration 2:  Log pseudolikelihood = -1.258e+08  (not concave)
Iteration 3:  Log pseudolikelihood =  -78258201  (not concave)
Iteration 4:  Log pseudolikelihood =  -64378107  (not concave)
Iteration 5:  Log pseudolikelihood =  -61543843  (not concave)
Iteration 6:  Log pseudolikelihood =  -60603310  (not concave)
Iteration 7:  Log pseudolikelihood =  -59883516  (not concave)
Iteration 8:  Log pseudolikelihood =  -59352578  (not concave)
Iteration 9:  Log pseudolikelihood =  -58966252  (not concave)
Iteration 10: Log pseudolikelihood =  -58703729  (not concave)
Iteration 11: Log pseudolikelihood =  -58495299  (not concave)
Iteration 12: Log pseudolikelihood =  -58341348  (not concave)
```

```
Iteration 13: Log pseudolikelihood =  -58219403  (not concave)
Iteration 14: Log pseudolikelihood =  -58124914  (not concave)
Iteration 15: Log pseudolikelihood =  -58046578
Iteration 16: Log pseudolikelihood =  -57485336
Iteration 17: Log pseudolikelihood =  -57248962  (not concave)
Iteration 18: Log pseudolikelihood =  -57198365
Iteration 19: Log pseudolikelihood =  -57176710
Iteration 20: Log pseudolikelihood =  -57174649
Iteration 21: Log pseudolikelihood =  -57174593
Iteration 22: Log pseudolikelihood =  -57174593

Stereotype logistic regression                          Number of obs =  22,899
                                                        Wald chi2(35) = 1185.54
Log pseudolikelihood = -57174593                        Prob > chi2   =  0.0000

 ( 1)  [phi1_1]_cons = 1
```

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.4622476	.1589527	2.91	0.004	.1507061	.7737891
c.age#c.age	-.0027637	.0010665	-2.59	0.010	-.004854	-.0006733
yedu	-.0987977	.0435408	-2.27	0.023	-.1841362	-.0134593
c.yedu#c.yedu	-.0004284	.0019643	-0.22	0.827	-.0042782	.0034215
thinc	-.5935118	.1335462	-4.44	0.000	-.8552575	-.331766
c.thinc#c.thinc	.1013752	.0630037	1.61	0.108	-.0221099	.2248602
gender						
Female	.0330977	.1235652	0.27	0.789	-.2090857	.275281
0.covid	0	(omitted)				
household						
Couple, both responding	.4208176	.1563622	2.69	0.007	.1143534	.7272818
Multiple, at least 1 responding	.4631724	.2349139	1.97	0.049	.0027496	.9235952
cjs						
Not applicable	1.776695	.800504	2.22	0.026	.2077365	3.345654
Retired	.9105351	.2911398	3.13	0.002	.3399117	1.481159
Unemployed	.4177737	.6737033	0.62	0.535	-.9026604	1.738208
Permanently sick or disabled	3.308429	.56144	5.89	0.000	2.208027	4.408831
Homemaker	1.09069	.3391018	3.22	0.001	.4260623	1.755317
Other	.0719336	.5366976	0.13	0.893	-.9799744	1.123842
gali						
Limited	5.066626	.1979047	25.60	0.000	4.678739	5.454512
lifesat	-.6071039	.1798876	-3.37	0.001	-.9596771	-.2545306

c.lifesat#c.lifesat	-.0096671	.0130116	-0.74	0.458	-.0351693 > .0158351
naly					
At least 1	-.7287381	.1611187	-4.52	0.000	-1.044525 > -.4129514
Not applicable	0	(omitted)			
rhfo					
At least 1	1.116279	.1972748	5.66	0.000	.7296272 > 1.50293
ghto					
At least 1	-.7831334	.1842902	-4.25	0.000	-1.144335 > -.4219314
Not applicable	-.1149398	.1419854	-0.81	0.418	-.393226 > .1633464
rhfo#ghto					
At least 1#At least 1	-.6925812	.4118434	-1.68	0.093	-1.49978 > .1146171
At least 1#Not applicable	-.0142814	.3482872	-0.04	0.967	-.6969119 > .668349
country					
Austria	.7643403	.2151592	3.55	0.000	.342636 > 1.186045
Germany	2.028285	.2543314	7.97	0.000	1.529805 > 2.526766
Netherlands	.5444088	.2261229	2.41	0.016	.1012162 > .9876015
Spain	2.403988	.2606052	9.22	0.000	1.893211 > 2.914765
Italy	1.784119	.2470408	7.22	0.000	1.299928 > 2.26831
France	1.753849	.2157895	8.13	0.000	1.33091 > 2.176789
Denmark	-.2162381	.2576436	-0.84	0.401	-.7212104 > .2887341
Switzerland	.3619699	.2412777	1.50	0.134	-.1109257 > .8348655
Belgium	.5555303	.207872	2.67	0.008	.1481086 > .9629519
Czech Republic	1.66726	.2727526	6.11	0.000	1.132675 > 2.201846
Slovenia	1.498033	.304387	4.92	0.000	.9014451 > 2.09462
/phi1_1	1	(constrained)			
/phi1_2	.8465055	.0262819	32.21	0.000	.7949939 > .8980171
/phi1_3	.6237852	.0197494	31.59	0.000	.5850771 > .6624933
/phi1_4	.3333102	.015535	21.46	0.000	.3028622 > .3637583
/phi1_5	0	(base outcome)			
/theta1	16.65908	5.885204	2.83	0.005	5.124297 > 28.19387
/theta2	15.47545	5.010435	3.09	0.002	5.65518 > 25.29573
/theta3	13.22991	3.714639	3.56	0.000	5.949354 > 20.51047
/theta4	7.834475	2.012465	3.89	0.000	3.890115 > 11.77883
/theta5	0	(base outcome)			

(sphus=Poor is the base outcome)
results_imp1_ste_13_5.xls

dir : seeout

Predictive margins Number of obs = **22,899**
Model VCE: **Robust**

1._predict: **Pr(sphus==Excellent), predict(pr outcome(#1))**
2._predict: **Pr(sphus==Very good), predict(pr outcome(#2))**
3._predict: **Pr(sphus==Good), predict(pr outcome(#3))**
4._predict: **Pr(sphus==Fair), predict(pr outcome(#4))**
5._predict: **Pr(sphus==Poor), predict(pr outcome(#5))**

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. interval]	
_predict#rhfo#ghto						
1#None#None	.0469014	.0022405	20.93	0.000	.0425101	.0512927
1#None#At least 1	.0600781	.0036425	16.49	0.000	.0529389	.0672172
1#None#Not applicable	.048689	.0025653	18.98	0.000	.0436611	.0537169
1#At least 1#None	.0319756	.0026718	11.97	0.000	.026739	.0372122
1#At least 1#At least 1	.0526567	.0057638	9.14	0.000	.0413597	.0639536
1#At least 1#Not applicable	.0334882	.0037209	9.00	0.000	.0261954	.040781
2#None#None	.1222651	.0038233	31.98	0.000	.1147716	.1297587
2#None#At least 1	.1428848	.0053568	26.67	0.000	.1323856	.1533839
2#None#Not applicable	.1252195	.0043758	28.62	0.000	.1166431	.1337959
2#At least 1#None	.0951676	.0050051	19.01	0.000	.0853578	.1049774
2#At least 1#At least 1	.1315912	.0091854	14.33	0.000	.1135882	.1495941
2#At least 1#Not applicable	.0981421	.0071718	13.68	0.000	.0840856	.1121986
3#None#None	.3923791	.0055367	70.87	0.000	.3815273	.4032309
3#None#At least 1	.4099336	.0065488	62.60	0.000	.3970982	.422769
3#None#Not applicable	.395256	.0058902	67.10	0.000	.3837114	.4068005
3#At least 1#None	.3596565	.0074213	48.46	0.000	.345111	.3742021
3#At least 1#At least 1	.4010395	.009136	43.90	0.000	.3831332	.4189458
3#At least 1#Not applicable	.3638533	.0102696	35.43	0.000	.3437253	.3839812
4#None#None	.3152283	.0053953	58.43	0.000	.3046537	.3258028
4#None#At least 1	.2909728	.0067663	43.00	0.000	.2777111	.3042346
4#None#Not applicable	.3118666	.0057294	54.43	0.000	.3006372	.3230959
4#At least 1#None	.3436701	.0073036	47.05	0.000	.3293553	.3579849
4#At least 1#At least 1	.3044783	.0110576	27.54	0.000	.2828057	.3261509
4#At least 1#Not applicable	.3407981	.0086684	39.31	0.000	.3238083	.3577879
5#None#None	.1232261	.0042766	28.81	0.000	.1148442	.1316081
5#None#At least 1	.0961308	.0059587	16.13	0.000	.084452	.1078095
5#None#Not applicable	.1189689	.0052049	22.86	0.000	.1087676	

```
> .1291703
          5#At least 1#None  |   .1695301   .0080996    20.93   0.000     .1536551
> .1854051
    5#At least 1#At least 1  |   .1102344   .0117161     9.41   0.000     .0872714
> .1331975
5#At least 1#Not applicable  |   .1637183   .0128401    12.75   0.000     .1385521
> .1888845
----------------------------+----------------------------------------------------
  ________
slog13_5.xls
dir : seeout
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -1.867e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.309e+08  (not concave)
Iteration 2:  Log pseudolikelihood =  -66791840  (not concave)
Iteration 3:  Log pseudolikelihood =  -52577582  (not concave)
Iteration 4:  Log pseudolikelihood =  -48715317  (not concave)
Iteration 5:  Log pseudolikelihood =  -48318738
Iteration 6:  Log pseudolikelihood =  -47338135  (not concave)
Iteration 7:  Log pseudolikelihood =  -47097065  (not concave)
Iteration 8:  Log pseudolikelihood =  -46901238  (not concave)
Iteration 9:  Log pseudolikelihood =  -46803329
Iteration 10: Log pseudolikelihood =  -46624014  (not concave)
Iteration 11: Log pseudolikelihood =  -46579552
Iteration 12: Log pseudolikelihood =  -46559185
Iteration 13: Log pseudolikelihood =  -46542867
Iteration 14: Log pseudolikelihood =  -46538183
Iteration 15: Log pseudolikelihood =  -46537624
Iteration 16: Log pseudolikelihood =  -46537624

Stereotype logistic regression                        Number of obs =  18,545
                                                      Wald chi2(32) = 1115.42
Log pseudolikelihood = -46537624                      Prob > chi2   =  0.0000

 ( 1)  [phi1_1]_cons = 1
----------------------------+----------------------------------------------------
  ____________
                            |               Robust
                      sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]
----------------------------+----------------------------------------------------
  ____________
                        age |   .6059445   .1712174     3.54   0.000     .2703646
>     .9415244
                            |
                 c.age#c.age |   -.003572   .0011191    -3.19   0.001    -.0057655
>    -.0013786
                            |
                       yedu |  -.0389072   .0468109    -0.83   0.406     -.130655
>     .0528405
                            |
              c.yedu#c.yedu |  -.0025157   .0021348    -1.18   0.239    -.0066998
>     .0016684
                            |
                      thinc |  -.5016595   .1559469    -3.22   0.001    -.8073099
>    -.1960092
                            |
            c.thinc#c.thinc |  -.0306653   .0722281    -0.42   0.671    -.1722298
>     .1108992
                            |
                     gender |
                     Female |    .049408   .1384878     0.36   0.721    -.2220231
>      .320839
                    0.covid |          0  (omitted)
                            |
                  household |
     Couple, both responding |    .399007   .1559476     2.56   0.011     .0933554
>     .7046586
Multiple, at least 1 responding |   .6504495   .2632122     2.47   0.013      .134563
>     1.166336
```

```
                                 |
                            cjs  |
     Not applicable  |   3.05226   .836656     3.65   0.000     1.412444
>     4.692075
                     Retired  |   1.540079  .4475196     3.44   0.001     .6629567
>     2.417201
                  Unemployed  |   .3671672   .853179     0.43   0.667    -1.305033
>     2.039367
   Permanently sick or disabled  |   3.287862  .7260471     4.53   0.000     1.864836
>     4.710888
                   Homemaker  |   1.776291  .4862919     3.65   0.000     .8231767
>     2.729406
                       Other  |   1.702932  .5402163     3.15   0.002     .6441278
>     2.761737
                                 |
                          gali  |
                     Limited  |   5.270265  .2051201    25.69   0.000     4.868237
>     5.672293
                      lifesat  |  -1.001323  .1854305    -5.40   0.000     -1.36476
>    -.6378859
                                 |
            c.lifesat#c.lifesat  |   .0149998  .0133065     1.13   0.260    -.0110804
>     .0410801
                                 |
                          naly  |
                  At least 1  |  -.3217266  .1587614    -2.03   0.043    -.6328933
>    -.0105599
     Not applicable  |          0  (omitted)
                                 |
                          rhfo  |
                  At least 1  |   1.312327  .1757359     7.47   0.000     .9678905
>     1.656763
                                 |
                          ghto  |
                  At least 1  |  -.7240298  .1929719    -3.75   0.000    -1.102248
>    -.3458118
                                 |
                      rhfo#ghto  |
        At least 1#At least 1  |  -1.089427  .3547504    -3.07   0.002    -1.784724
>    -.3941286
                                 |
                       country  |
                     Austria  |   .4788158  .2158005     2.22   0.027     .0558547
>     .9017769
                     Germany  |   1.777202  .2540067     7.00   0.000     1.279358
>     2.275046
                       Spain  |    2.14522  .2610277     8.22   0.000     1.633615
>     2.656825
                       Italy  |   1.645484  .2453655     6.71   0.000     1.164576
>     2.126391
                      France  |   1.577455  .2180781     7.23   0.000      1.15003
>      2.00488
                     Denmark  |  -.5171752   .272806    -1.90   0.058    -1.051865
>     .0175148
                 Switzerland  |   .3615547  .2564153     1.41   0.159      -.14101
>     .8641194
                     Belgium  |   .2812452  .2112397     1.33   0.183    -.1327769
>     .6952674
               Czech Republic  |   1.745704   .275224     6.34   0.000     1.206275
>     2.285134
                    Slovenia  |   1.849635  .3060507     6.04   0.000     1.249786
>     2.449483
-------------------------------+----------------------------------------------------------------
                     /phi1_1  |          1  (constrained)
                     /phi1_2  |   .8814561  .0253219    34.81   0.000      .831826
>     .9310861
                     /phi1_3  |   .6605047  .0193769    34.09   0.000     .6225266
>     .6984828
                     /phi1_4  |   .3549683  .0165504    21.45   0.000     .3225301
>     .3874065
                     /phi1_5  |          0  (base outcome)
```

```
------------------------------------------------------------------------------
             /theta1 |   22.82744   6.550569     3.48   0.000     9.988561
>     35.66632
             /theta2 |   21.54047   5.765991     3.74   0.000     10.23933
>      32.8416
             /theta3 |   17.98445   4.350802     4.13   0.000     9.457035
>     26.51186
             /theta4 |   10.54764   2.405219     4.39   0.000       5.8335
>     15.26178
             /theta5 |          0  (base outcome)
------------------------------------------------------------------------------
(sphus=Poor is the base outcome)
results_imp1_ste_15_5.xls
dir : seeout

Predictive margins                                      Number of obs = 18,545
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------------------------------------------------------
                             |            Delta-method
                             |     Margin   std. err.      z    P>|z|     [95% conf. inter
> val]
-----------------------------+------------------------------------------------------------
  _predict#rhfo#ghto         |
        1#None#None          |   .0423035   .0024071    17.57   0.000     .0375858    .047
> 0213
      1#None#At least 1      |   .0523654   .0033276    15.74   0.000     .0458433    .058
> 8874
      1#At least 1#None      |   .0277424   .0023607    11.75   0.000     .0231154    .032
> 3694
1#At least 1#At least 1      |   .0491049   .0046506    10.56   0.000       .03999    .058
> 2199
        2#None#None          |   .1242797   .0040609    30.60   0.000     .1163204    .132
> 2389
      2#None#At least 1      |   .1442243   .0061009    23.64   0.000     .1322668    .156
> 1818
      2#At least 1#None      |   .0917211   .0048698    18.83   0.000     .0821765    .101
> 2657
2#At least 1#At least 1      |   .1379521   .0080637    17.11   0.000     .1221475    .153
> 7566
        3#None#None          |   .3881645   .0062442    62.16   0.000     .3759261     .40
> 0403
      3#None#At least 1      |    .406641   .0079436    51.19   0.000     .3910719    .422
> 2102
      3#At least 1#None      |   .3460833   .0070133    49.35   0.000     .3323374    .359
> 8291
3#At least 1#At least 1      |   .4013616   .0089135    45.03   0.000     .3838913    .418
> 8318
        4#None#None          |   .3318642   .0058426    56.80   0.000     .3204129    .343
> 3155
      4#None#At least 1      |   .3079906   .0078815    39.08   0.000     .2925433     .32
> 3438
      4#At least 1#None      |   .3659816   .0074805    48.92   0.000     .3513201    .380
> 6431
4#At least 1#At least 1      |   .3156649   .0102774    30.71   0.000     .2955215    .335
> 8083
        5#None#None          |    .113388   .0042166    26.89   0.000     .1051238    .121
> 6523
      5#None#At least 1      |   .0887786   .0063335    14.02   0.000     .0763651    .101
> 1921
      5#At least 1#None      |   .1684716   .0074289    22.68   0.000     .1539112    .183
> 0321
```

```
5#At least 1#At least 1  |   .0959165   .0090822    10.56   0.000     .0781157    .113
> 7173
-------------------------+----------------------------------------------------------------
  ____
slog15_5.xls
dir : seeout
note: 0.covid omitted because of collinearity.
note: 2.naly omitted because of collinearity.
note: 2.ghto omitted because of collinearity.
note: 0.rhfo#2.ghto identifies no observations in the sample.
note: 1.rhfo#2.ghto identifies no observations in the sample.
note: 2.rhfo#0.ghto identifies no observations in the sample.
note: 2.rhfo#1.ghto identifies no observations in the sample.
note: 2.rhfo#2.ghto omitted because of collinearity.

Iteration 0:  Log pseudolikelihood = -1.981e+08  (not concave)
Iteration 1:  Log pseudolikelihood =  -94024603  (not concave)
Iteration 2:  Log pseudolikelihood =  -59643469  (not concave)
Iteration 3:  Log pseudolikelihood =  -46282613  (not concave)
Iteration 4:  Log pseudolikelihood =  -42680562  (not concave)
Iteration 5:  Log pseudolikelihood =  -42309239  (not concave)
Iteration 6:  Log pseudolikelihood =  -42082424  (not concave)
Iteration 7:  Log pseudolikelihood =  -41991306
Iteration 8:  Log pseudolikelihood =  -41880693
Iteration 9:  Log pseudolikelihood =  -41841555  (not concave)
Iteration 10: Log pseudolikelihood =  -41826829
Iteration 11: Log pseudolikelihood =  -41817671  (not concave)
Iteration 12: Log pseudolikelihood =  -41784002
Iteration 13: Log pseudolikelihood =  -41778723
Iteration 14: Log pseudolikelihood =  -41777600
Iteration 15: Log pseudolikelihood =  -41777568
Iteration 16: Log pseudolikelihood =  -41777568

Stereotype logistic regression                          Number of obs = 16,181
                                                        Wald chi2(33) = 853.07
Log pseudolikelihood = -41777568                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
------------------------------------------------------------------------------------------
  ____________
                             |               Robust
                       sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]                |
------------------------------------------------------------------------------------------
  ____________
                         age |   .2440029   .1807981     1.35   0.177    -.1103549
>     .5983607
                             |
                 c.age#c.age |   -.001215   .0011688    -1.04   0.299    -.0035059
>     .0010758
                             |
                        yedu |  -.0289619   .0466388    -0.62   0.535    -.1203724
>     .0624485
                             |
               c.yedu#c.yedu |  -.0032029   .0021066    -1.52   0.128    -.0073318
>     .0009261
                             |
                       thinc |  -.1836634   .1627623    -1.13   0.259    -.5026717
>     .1353449
                             |
             c.thinc#c.thinc |  -.1894622   .0879717    -2.15   0.031    -.3618835
>    -.0170409
                             |
                      gender |
                      Female |  -.2070349   .1367634    -1.51   0.130    -.4750862
>     .0610164
                     0.covid |          0  (omitted)
                             |
                   household |
        Couple, both responding |   .1075268    .148467     0.72   0.469    -.1834631
>     .3985168
Multiple, at least 1 responding |   .1553773    .271861     0.57   0.568    -.3774604
```

```
>     .6882149
                                  |
                             cjs  |
                 Not applicable   |   3.070105   .7067026     4.34   0.000     1.684993
>     4.455216
                         Retired  |   1.258456   .5474555     2.30   0.022     .1854628
>     2.331449
                      Unemployed  |   .9416258   1.736951     0.54   0.588    -2.462735
>     4.345987
   Permanently sick or disabled   |   4.038788   .8660498     4.66   0.000     2.341361
>     5.736214
                       Homemaker  |   1.535448   .5807254     2.64   0.008     .3972476
>     2.673649
                           Other  |   1.032989   .7341741     1.41   0.159    -.4059658
>     2.471944
                                  |
                            gali  |
                         Limited  |   4.772781   .2106237    22.66   0.000     4.359966
>     5.185595
                          lifesat |  -.2996385   .1978931    -1.51   0.130    -.6875018
>     .0882248
                                  |
             c.lifesat#c.lifesat  |  -.0258529   .0141972    -1.82   0.069    -.0536789
>     .0019731
                                  |
                            naly  |
                       At least 1 |  -.3586821   .1659821    -2.16   0.031     -.684001
>    -.0333632
                 Not applicable   |          0  (omitted)
                                  |
                            rhfo  |
                       At least 1 |   .8231215     .23117     3.56   0.000     .3700367
>     1.276206
                 Not applicable   |   .2982721   .2508554     1.19   0.234    -.1933954
>     .7899395
                                  |
                            ghto  |
                       At least 1 |  -.6870735   .2327447    -2.95   0.003    -1.143245
>    -.2309022
                 Not applicable   |          0  (omitted)
                                  |
                        rhfo#ghto |
            None#Not applicable   |          0  (empty)
          At least 1#At least 1   |   .7583427   .4243732     1.79   0.074    -.0734135
>     1.590099
      At least 1#Not applicable   |          0  (empty)
            Not applicable#None   |          0  (empty)
      Not applicable#At least 1   |          0  (empty)
  Not applicable#Not applicable   |          0  (omitted)
                                  |
                         country  |
                         Austria  |   .6890928   .2250001     3.06   0.002     .2481007
>     1.130085
                         Germany  |   1.576157   .2632995     5.99   0.000     1.060099
>     2.092214
                           Spain  |   2.145663   .2675139     8.02   0.000     1.621345
>      2.66998
                           Italy  |   1.606909   .2485869     6.46   0.000     1.119687
>      2.09413
                          France  |   1.105781   .2223345     4.97   0.000     .6700133
>     1.541548
                         Denmark  |  -.7656618   .2675486    -2.86   0.004    -1.290047
>    -.2412761
                     Switzerland  |   .0623345    .236451     0.26   0.792    -.4011009
>       .52577
                         Belgium  |   .0330348   .2104908     0.16   0.875    -.3795196
>     .4455892
                  Czech Republic  |   .3037219   .2395896     1.27   0.205    -.1658651
>      .773309
                        Slovenia  |   1.596388    .282196     5.66   0.000     1.043293
>     2.149482
------------------------------------+----------------------------------------------------------
```

```
  ____________
                        /phi1_1 |          1  (constrained)
                        /phi1_2 |   .920819    .031445    29.28   0.000      .859188
>       .98245
                        /phi1_3 |  .6713219   .0230077    29.18   0.000     .6262276
>     .7164162
                        /phi1_4 |  .3459023   .0196039    17.64   0.000     .3074793
>     .3843252
                        /phi1_5 |          0  (base outcome)
------------------------------------------------------------------------------
  ____________
                        /theta1 |  10.81587   7.016176     1.54   0.123    -2.935583
>     24.56732
                        /theta2 |  11.22374   6.485057     1.73   0.084    -1.486735
>     23.93422
                        /theta3 |  9.981565   4.737708     2.11   0.035     .6958282
>      19.2673
                        /theta4 |  5.956583    2.48332     2.40   0.016     1.089365
>      10.8238
                        /theta5 |          0  (base outcome)
------------------------------------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_17_5.xls
dir : seeout

Predictive margins                                      Number of obs = 16,181
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------------------------------------------------------
  ____________
                                |            Delta-method
                                |     Margin   std. err.      z    P>|z|     [95% con
> f. interval]
------------------------------------------------------------------------------
  ____________
             _predict#rhfo#ghto |
                   1#None#None  |  .0406955    .002633    15.46   0.000     .0355349
>     .0458561
             1#None#At least 1  |   .049934   .0039637    12.60   0.000     .0421652
>     .0577028
          1#None#Not applicable |          .  (not estimable)
             1#At least 1#None  |  .031251   .0028747    10.87   0.000     .0256167
>     .0368852
        1#At least 1#At least 1 |  .0305134   .0036946     8.26   0.000      .023272
>     .0377548
    1#At least 1#Not applicable |          .  (not estimable)
          1#Not applicable#None |          .  (not estimable)
    1#Not applicable#At least 1 |          .  (not estimable)
1#Not applicable#Not applicable |  .0370731   .0028918    12.82   0.000     .0314052
>      .042741
                   2#None#None  |  .1206207   .0051263    23.53   0.000     .1105733
>     .1306681
             2#None#At least 1  |  .1420561   .0082333    17.25   0.000     .1259192
>      .158193
          2#None#Not applicable |          .  (not estimable)
             2#At least 1#None  |  .0973274   .0065221    14.92   0.000     .0845443
>     .1101105
        2#At least 1#At least 1 |  .0954403   .0084887    11.24   0.000     .0788028
>     .1120778
    2#At least 1#Not applicable |          .  (not estimable)
          2#Not applicable#None |          .  (not estimable)
    2#Not applicable#At least 1 |          .  (not estimable)
2#Not applicable#Not applicable |  .1118672    .006276    17.82   0.000     .0995665
>     .1241679
                   3#None#None  |  .3839397   .0073406    52.30   0.000     .3695523
```

```
>     .3983271
              3#None#At least 1 |    .403913   .0094803    42.61   0.000      .385332
>      .422494
         3#None#Not applicable |          .  (not estimable)
              3#At least 1#None |   .3552339   .0093804    37.87   0.000     .3368487
>     .3736191
        3#At least 1#At least 1 |   .3525374   .0132853    26.54   0.000     .3264987
>     .3785762
    3#At least 1#Not applicable |          .  (not estimable)
         3#Not applicable#None |          .  (not estimable)
    3#Not applicable#At least 1 |          .  (not estimable)
3#Not applicable#Not applicable |   .3740883   .0080139    46.68   0.000     .3583813
>     .3897954
                    4#None#None |   .3311897   .0071282    46.46   0.000     .3172187
>     .3451607
              4#None#At least 1 |   .3066642   .0103418    29.65   0.000     .2863946
>     .3269337
         4#None#Not applicable |          .  (not estimable)
              4#At least 1#None |   .3559646   .0086246    41.27   0.000     .3390607
>     .3728685
        4#At least 1#At least 1 |   .3578407   .0104116    34.37   0.000     .3374343
>      .378247
    4#At least 1#Not applicable |          .  (not estimable)
         4#Not applicable#None |          .  (not estimable)
    4#Not applicable#At least 1 |          .  (not estimable)
4#Not applicable#Not applicable |   .3408008   .0078172    43.60   0.000     .3254794
>     .3561222
                    5#None#None |   .1235544   .0058974    20.95   0.000     .1119958
>     .1351131
              5#None#At least 1 |   .0974328    .008552    11.39   0.000     .0806711
>     .1141944
         5#None#Not applicable |          .  (not estimable)
              5#At least 1#None |   .1602231   .0104765    15.29   0.000     .1396896
>     .1807567
        5#At least 1#At least 1 |   .1636682   .0155924    10.50   0.000     .1331076
>     .1942289
    5#At least 1#Not applicable |          .  (not estimable)
         5#Not applicable#None |          .  (not estimable)
    5#Not applicable#At least 1 |          .  (not estimable)
5#Not applicable#Not applicable |   .1361706   .0084029    16.21   0.000     .1197012
>     .1526401
------------------------------+------------------------------------------------------------
```

slog17_5.xls
dir : seeout
note: **0.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -2.242e+08  (not concave)
Iteration 1:  Log pseudolikelihood =  -99865903  (not concave)
Iteration 2:  Log pseudolikelihood =  -46077189  (not concave)
Iteration 3:  Log pseudolikelihood =  -37240177  (not concave)
Iteration 4:  Log pseudolikelihood =  -32744791  (not concave)
Iteration 5:  Log pseudolikelihood =  -30573470  (not concave)
Iteration 6:  Log pseudolikelihood =  -30087051  (not concave)
Iteration 7:  Log pseudolikelihood =  -29725259  (not concave)
Iteration 8:  Log pseudolikelihood =  -29594879  (not concave)
Iteration 9:  Log pseudolikelihood =  -29510386  (not concave)
Iteration 10: Log pseudolikelihood =  -29434797  (not concave)
Iteration 11: Log pseudolikelihood =  -29339858  (not concave)
Iteration 12: Log pseudolikelihood =  -29221102  (not concave)
Iteration 13: Log pseudolikelihood =  -29161794  (not concave)
Iteration 14: Log pseudolikelihood =  -29051840  (not concave)
Iteration 15: Log pseudolikelihood =  -28962142  (not concave)
Iteration 16: Log pseudolikelihood =  -28874976  (not concave)
Iteration 17: Log pseudolikelihood =  -28830374  (not concave)
Iteration 18: Log pseudolikelihood =  -28765167  (not concave)
Iteration 19: Log pseudolikelihood =  -28725740  (not concave)
Iteration 20: Log pseudolikelihood =  -28705580  (not concave)
Iteration 21: Log pseudolikelihood =  -28596349  (not concave)
Iteration 22: Log pseudolikelihood =  -28535848  (not concave)
Iteration 23: Log pseudolikelihood =  -28505966  (not concave)
```

```
Iteration 24: Log pseudolikelihood =  -28491679  (not concave)
Iteration 25: Log pseudolikelihood =  -28481447  (not concave)
Iteration 26: Log pseudolikelihood =  -28471743  (not concave)
Iteration 27: Log pseudolikelihood =  -28459988  (not concave)
Iteration 28: Log pseudolikelihood =  -28442425  (not concave)
Iteration 29: Log pseudolikelihood =  -28433855  (not concave)
Iteration 30: Log pseudolikelihood =  -28426649  (not concave)
Iteration 31: Log pseudolikelihood =  -28419595  (not concave)
Iteration 32: Log pseudolikelihood =  -28412674  (not concave)
Iteration 33: Log pseudolikelihood =  -28405747  (not concave)
Iteration 34: Log pseudolikelihood =  -28398843  (not concave)
Iteration 35: Log pseudolikelihood =  -28391995  (not concave)
Iteration 36: Log pseudolikelihood =  -28385188  (not concave)
Iteration 37: Log pseudolikelihood =  -28378397  (not concave)
Iteration 38: Log pseudolikelihood =  -28371641  (not concave)
Iteration 39: Log pseudolikelihood =  -28364915  (not concave)
Iteration 40: Log pseudolikelihood =  -28358221  (not concave)
Iteration 41: Log pseudolikelihood =  -28351547  (not concave)
Iteration 42: Log pseudolikelihood =  -28344906  (not concave)
Iteration 43: Log pseudolikelihood =  -28338289  (not concave)
Iteration 44: Log pseudolikelihood =  -28331703  (not concave)
Iteration 45: Log pseudolikelihood =  -28325139  (not concave)
Iteration 46: Log pseudolikelihood =  -28318608  (not concave)
Iteration 47: Log pseudolikelihood =  -28312100  (not concave)
Iteration 48: Log pseudolikelihood =  -28305625  (not concave)
Iteration 49: Log pseudolikelihood =  -28299174  (not concave)
Iteration 50: Log pseudolikelihood =  -28292756  (not concave)
Iteration 51: Log pseudolikelihood =  -28286364  (not concave)
Iteration 52: Log pseudolikelihood =  -28280008  (not concave)
Iteration 53: Log pseudolikelihood =  -28273678  (not concave)
Iteration 54: Log pseudolikelihood =  -28267384  (not concave)
Iteration 55: Log pseudolikelihood =  -28261119  (not concave)
Iteration 56: Log pseudolikelihood =  -28254892  (not concave)
Iteration 57: Log pseudolikelihood =  -28248695  (not concave)
Iteration 58: Log pseudolikelihood =  -28242538  (not concave)
Iteration 59: Log pseudolikelihood =  -28236413  (not concave)
Iteration 60: Log pseudolikelihood =  -28230329  (not concave)
Iteration 61: Log pseudolikelihood =  -28224280  (not concave)
Iteration 62: Log pseudolikelihood =  -28218274  (not concave)
Iteration 63: Log pseudolikelihood =  -28212303  (not concave)
Iteration 64: Log pseudolikelihood =  -28206378  (not concave)
Iteration 65: Log pseudolikelihood =  -28200491  (not concave)
Iteration 66: Log pseudolikelihood =  -28194651  (not concave)
Iteration 67: Log pseudolikelihood =  -28188851  (not concave)
Iteration 68: Log pseudolikelihood =  -28183100  (not concave)
Iteration 69: Log pseudolikelihood =  -28177391  (not concave)
Iteration 70: Log pseudolikelihood =  -28171733  (not concave)
Iteration 71: Log pseudolikelihood =  -28166119  (not concave)
Iteration 72: Log pseudolikelihood =  -28160556  (not concave)
Iteration 73: Log pseudolikelihood =  -28155040  (not concave)
Iteration 74: Log pseudolikelihood =  -28149578
Iteration 75: Log pseudolikelihood =  -27959856  (not concave)
Iteration 76: Log pseudolikelihood =  -27889583
Iteration 77: Log pseudolikelihood =  -27884001
Iteration 78: Log pseudolikelihood =  -27859949
Iteration 79: Log pseudolikelihood =  -27850381  (not concave)
Iteration 80: Log pseudolikelihood =  -27841465
Iteration 81: Log pseudolikelihood =  -27839460
Iteration 82: Log pseudolikelihood =  -27839359
Iteration 83: Log pseudolikelihood =  -27839358
Iteration 84: Log pseudolikelihood =  -27839358

Stereotype logistic regression                          Number of obs = 10,710
                                                        Wald chi2(33) =  760.30
Log pseudolikelihood = -27839358                        Prob > chi2   =  0.0000
```

(1) **[phi1_1]_cons = 1**

sphus	Coefficient	Robust std. err.	z	P>|z|	[95% conf. interval]	
age	.5859297	.2535367	2.31	0.021	.0890069	1.082853
c.age#c.age	-.0034512	.0015856	-2.18	0.030	-.0065589	-.0003436
yedu	-.0190873	.060603	-0.31	0.753	-.137867	.0996925
c.yedu#c.yedu	-.0028336	.002716	-1.04	0.297	-.0081569	.0024896
thinc	-.2553817	.1859417	-1.37	0.170	-.6198209	.1090574
c.thinc#c.thinc	-.0197741	.0898322	-0.22	0.826	-.1958419	.1562938
gender						
Female	-.0059373	.1794588	-0.03	0.974	-.3576701	.3457956
0.covid	0	(omitted)				
household						
Couple, both responding	.5709996	.205611	2.78	0.005	.1680094	.9739898
Multiple, at least 1 responding	.5948981	.3473435	1.71	0.087	-.0858826	1.275679
cjs						
Not applicable	3.751262	1.283411	2.92	0.003	1.235822	6.266702
Retired	3.773043	1.098556	3.43	0.001	1.619912	5.926174
Unemployed	2.847266	2.544588	1.12	0.263	-2.140035	7.834567
Permanently sick or disabled	5.196977	1.354937	3.84	0.000	2.541349	7.852605
Homemaker	3.356952	1.126924	2.98	0.003	1.148222	5.565683
Other	3.569468	1.212503	2.94	0.003	1.193005	5.94593
gali						
Limited	5.290246	.2886749	18.33	0.000	4.724454	5.856039
lifesat	-1.39156	.3163785	-4.40	0.000	-2.01165	-.7714694
c.lifesat#c.lifesat	.0395427	.022099	1.79	0.074	-.0037704	.0828559
naly						
At least 1	-1.015524	.2636063	-3.85	0.000	-1.532183	-.4988655
Not applicable	0	(omitted)				
rhfo						
At least 1	.8397092	.2264292	3.71	0.000	.395916	1.283502
ghto						
At least 1	-.5736995	.2407795	-2.38	0.017	-1.045619	-.1017805

```
                                 |
                      rhfo#ghto  |
         At least 1#At least 1   |   -.276062   .4689503    -0.59   0.556    -1.195188
>     .6430638
                                 |
                        country  |
                        Austria  |   .4933457   .3020781     1.63   0.102    -.0987165
>     1.085408
                        Germany  |   1.112176   .3403397     3.27   0.001     .4451229
>      1.77923
                    Netherlands  |  -.1896636   .3218755    -0.59   0.556    -.8205281
>     .4412009
                          Spain  |   2.107817   .3638614     5.79   0.000     1.394662
>     2.820972
                          Italy  |   1.235478   .3602867     3.43   0.001      .529329
>     1.941627
                         France  |   .9581815   .2879369     3.33   0.001     .3938356
>     1.522527
                        Denmark  |  -.4209875   .3503556    -1.20   0.230    -1.107672
>     .2656969
                    Switzerland  |  -.0832695   .3366261    -0.25   0.805    -.7430445
>     .5765055
                        Belgium  |   .0365054   .2980786     0.12   0.903     -.547718
>     .6207287
                  Czech Republic |  -.3227373   .3430341    -0.94   0.347    -.9950719
>     .3495973
                       Slovenia  |   1.135888   .3780788     3.00   0.003     .3948674
>     1.876909
-------------------------------------------------------------------------------------
                        /phi1_1  |          1  (constrained)
                        /phi1_2  |   .8497599   .0369926    22.97   0.000     .7772556
>     .9222641
                        /phi1_3  |   .6514199   .0276262    23.58   0.000     .5972736
>     .7055662
                        /phi1_4  |   .3400751   .0229491    14.82   0.000     .2950957
>     .3850545
                        /phi1_5  |          0  (base outcome)
-------------------------------------------------------------------------------------
                        /theta1  |   21.89902   10.09401     2.17   0.030     2.115125
>     41.68292
                        /theta2  |    20.2677   8.711816     2.33   0.020      3.19285
>     37.34254
                        /theta3  |    17.2674   6.646709     2.60   0.009     4.240087
>     30.29471
                        /theta4  |   9.948545   3.554801     2.80   0.005     2.981263
>     16.91583
                        /theta5  |          0  (base outcome)
-------------------------------------------------------------------------------------
```

(sphus=Poor is the base outcome)
results_imp1_ste_20_5.xls
dir : seeout

Predictive margins Number of obs = **10,710**
Model VCE: **Robust**

1._predict: **Pr(sphus==Excellent), predict(pr outcome(#1))**
2._predict: **Pr(sphus==Very good), predict(pr outcome(#2))**
3._predict: **Pr(sphus==Good), predict(pr outcome(#3))**
4._predict: **Pr(sphus==Fair), predict(pr outcome(#4))**
5._predict: **Pr(sphus==Poor), predict(pr outcome(#5))**

	Margin	Delta-method std. err.	z	P>|z|	[95% conf. interval]	
_predict#rhfo#ghto						
1#None#None	.0365597	.0026983	13.55	0.000	.0312711	.0418483
1#None#At least 1	.0437498	.0038015	11.51	0.000	.0362991	.0512005
1#At least 1#None	.0276913	.0027814	9.96	0.000	.0222399	.0331427
1#At least 1#At least 1	.0366774	.0049724	7.38	0.000	.0269316	.0464232
2#None#None	.129767	.0058419	22.21	0.000	.118317	.1412169
2#None#At least 1	.1455365	.0078903	18.45	0.000	.1300719	.1610011
2#At least 1#None	.1081513	.0064457	16.78	0.000	.095518	.1207847
2#At least 1#At least 1	.1300367	.0110086	11.81	0.000	.1084604	.1516131
3#None#None	.387978	.0081525	47.59	0.000	.3719994	.4039565
3#None#At least 1	.4045002	.0104717	38.63	0.000	.383976	.4250245
3#At least 1#None	.3600503	.0095665	37.64	0.000	.3413004	.3788003
3#At least 1#At least 1	.3882868	.0139118	27.91	0.000	.3610201	.4155535
4#None#None	.3413868	.0078356	43.57	0.000	.3260294	.3567443
4#None#At least 1	.32035	.0107096	29.91	0.000	.2993595	.3413404
4#At least 1#None	.3680916	.0098638	37.32	0.000	.348759	.3874243
4#At least 1#At least 1	.3410359	.0145194	23.49	0.000	.3125785	.3694934
5#None#None	.1043085	.0054031	19.31	0.000	.0937186	.1148984
5#None#At least 1	.0858635	.0074931	11.46	0.000	.0711773	.1005498
5#At least 1#None	.1360154	.0084853	16.03	0.000	.1193846	.1526462
5#At least 1#At least 1	.1039631	.0136534	7.61	0.000	.0772029	.1307233

slog20_5.xls
dir : seeout
note: **1.covid** omitted because of collinearity.
note: **2.naly** omitted because of collinearity.

```
Iteration 0:  Log pseudolikelihood = -1.773e+08  (not concave)
Iteration 1:  Log pseudolikelihood = -1.490e+08  (not concave)
Iteration 2:  Log pseudolikelihood =  -95794005  (not concave)
Iteration 3:  Log pseudolikelihood =  -78795750  (not concave)
Iteration 4:  Log pseudolikelihood =  -56984866  (not concave)
Iteration 5:  Log pseudolikelihood =  -52059667  (not concave)
Iteration 6:  Log pseudolikelihood =  -40816577  (not concave)
Iteration 7:  Log pseudolikelihood =  -37582863  (not concave)
Iteration 8:  Log pseudolikelihood =  -35956294  (not concave)
Iteration 9:  Log pseudolikelihood =  -33672176  (not concave)
Iteration 10: Log pseudolikelihood =  -32545329  (not concave)
Iteration 11: Log pseudolikelihood =  -30818662  (not concave)
Iteration 12: Log pseudolikelihood =  -30448932  (not concave)
Iteration 13: Log pseudolikelihood =  -30173556  (not concave)
Iteration 14: Log pseudolikelihood =  -30030114  (not concave)
Iteration 15: Log pseudolikelihood =  -29920555  (not concave)
Iteration 16: Log pseudolikelihood =  -29841059  (not concave)
```

```
Iteration 17: Log pseudolikelihood =  -29800002  (not concave)
Iteration 18: Log pseudolikelihood =  -29738065  (not concave)
Iteration 19: Log pseudolikelihood =  -29681862  (not concave)
Iteration 20: Log pseudolikelihood =  -29381861  (not concave)
Iteration 21: Log pseudolikelihood =  -29261063  (not concave)
Iteration 22: Log pseudolikelihood =  -29223381  (not concave)
Iteration 23: Log pseudolikelihood =  -29185409  (not concave)
Iteration 24: Log pseudolikelihood =  -29037528  (not concave)
Iteration 25: Log pseudolikelihood =  -28973304  (not concave)
Iteration 26: Log pseudolikelihood =  -28943877  (not concave)
Iteration 27: Log pseudolikelihood =  -28921579  (not concave)
Iteration 28: Log pseudolikelihood =  -28902768  (not concave)
Iteration 29: Log pseudolikelihood =  -28884530  (not concave)
Iteration 30: Log pseudolikelihood =  -28864234  (not concave)
Iteration 31: Log pseudolikelihood =  -28830904  (not concave)
Iteration 32: Log pseudolikelihood =  -28795411  (not concave)
Iteration 33: Log pseudolikelihood =  -28774162  (not concave)
Iteration 34: Log pseudolikelihood =  -28756129  (not concave)
Iteration 35: Log pseudolikelihood =  -28737484  (not concave)
Iteration 36: Log pseudolikelihood =  -28719991  (not concave)
Iteration 37: Log pseudolikelihood =  -28702934  (not concave)
Iteration 38: Log pseudolikelihood =  -28685473  (not concave)
Iteration 39: Log pseudolikelihood =  -28668370  (not concave)
Iteration 40: Log pseudolikelihood =  -28651729  (not concave)
Iteration 41: Log pseudolikelihood =  -28635276  (not concave)
Iteration 42: Log pseudolikelihood =  -28618881  (not concave)
Iteration 43: Log pseudolikelihood =  -28602840  (not concave)
Iteration 44: Log pseudolikelihood =  -28587086  (not concave)
Iteration 45: Log pseudolikelihood =  -28571624  (not concave)
Iteration 46: Log pseudolikelihood =  -28556377  (not concave)
Iteration 47: Log pseudolikelihood =  -28541485  (not concave)
Iteration 48: Log pseudolikelihood =  -28526876  (not concave)
Iteration 49: Log pseudolikelihood =  -28512616  (not concave)
Iteration 50: Log pseudolikelihood =  -28498639  (not concave)
Iteration 51: Log pseudolikelihood =  -28485033
Iteration 52: Log pseudolikelihood =  -28390030  (not concave)
Iteration 53: Log pseudolikelihood =  -28273932  (not concave)
Iteration 54: Log pseudolikelihood =  -28258669
Iteration 55: Log pseudolikelihood =  -28166816  (not concave)
Iteration 56: Log pseudolikelihood =  -28143626
Iteration 57: Log pseudolikelihood =  -28133309
Iteration 58: Log pseudolikelihood =  -28127188
Iteration 59: Log pseudolikelihood =  -28127061
Iteration 60: Log pseudolikelihood =  -28127060
Iteration 61: Log pseudolikelihood =  -28127060

Stereotype logistic regression                          Number of obs = 11,204
                                                        Wald chi2(32) = 589.52
Log pseudolikelihood = -28127060                        Prob > chi2   = 0.0000

 ( 1)  [phi1_1]_cons = 1
------------------------------------------------------------------------------
             |               Robust
       sphus | Coefficient  std. err.      z    P>|z|     [95% con
> f. interval]
-------------+----------------------------------------------------------------
         age |   .9176221   .3072751     2.99   0.003      .315374
>      1.51987
             |
   c.age#c.age |  -.0053309   .0018971    -2.81   0.005    -.0090492
>    -.0016126
             |
        yedu |   .0348572   .0666843     0.52   0.601    -.0958416
>     .1655561
             |
 c.yedu#c.yedu |  -.0045109   .0029379    -1.54   0.125    -.0102691
>     .0012472
             |
       thinc |  -.1872652   .1992277    -0.94   0.347    -.5777443
>     .2032139
```

```
                                 |
                c.thinc#c.thinc  |   -.076148   .0946057    -0.80   0.421    -.2615717
>     .1092757
                                 |
                         gender  |
                         Female  |   .1161801    .182866     0.64   0.525    -.2422306
>     .4745909
                        1.covid  |          0  (omitted)
                                 |
                      household  |
        Couple, both responding  |   .6568177   .2075926     3.16   0.002     .2499437
>     1.063692
Multiple, at least 1 responding  |   .3370498   .4043052     0.83   0.404    -.4553738
>     1.129473
                                 |
                            cjs  |
                Not applicable   |  -.5100745   1.962304    -0.26   0.795    -4.356119
>      3.33597
                        Retired  |   -.304071   1.836633    -0.17   0.869    -3.903806
>     3.295664
   Permanently sick or disabled  |   .8453435   2.135802     0.40   0.692    -3.340751
>     5.031438
                      Homemaker  |  -.7388283   1.858905    -0.40   0.691    -4.382215
>     2.904559
                          Other  |   .1184317   1.939617     0.06   0.951    -3.683148
>     3.920012
                                 |
                           gali  |
                        Limited  |   5.689251   .3074098    18.51   0.000     5.086739
>     6.291763
                         lifesat |  -.9700287   .2863441    -3.39   0.001    -1.531253
>    -.4088045
                                 |
             c.lifesat#c.lifesat |   .0013629   .0201907     0.07   0.946    -.0382101
>     .0409359
                                 |
                           naly  |
                       At least 1 |  -.9300217   .2295964    -4.05   0.000    -1.380022
>     -.480021
                Not applicable   |          0  (omitted)
                                 |
                           rhfo  |
                       At least 1 |   .8210704   .2223303     3.69   0.000     .3853109
>      1.25683
                                 |
                           ghto  |
                       At least 1 |  -.6879999   .2838318    -2.42   0.015      -1.2443
>    -.1316998
                                 |
                       rhfo#ghto |
          At least 1#At least 1  |   .1725653   .5076238     0.34   0.734    -.8223591
>      1.16749
                                 |
                         country |
                         Austria |   .9794453   .3076506     3.18   0.001     .3764611
>      1.58243
                         Germany |   1.646007   .3662123     4.49   0.000     .9282445
>      2.36377
                     Netherlands |  -.0858809    .342887    -0.25   0.802     -.757927
>     .5861652
                           Spain |   2.308087   .3812135     6.05   0.000     1.560922
>     3.055252
                           Italy |   1.744485   .3575028     4.88   0.000     1.043793
>     2.445178
                          France |   1.113653   .3076903     3.62   0.000      .510591
>     1.716715
                         Denmark |  -.0705074    .367239    -0.19   0.848    -.7902827
>     .6492679
                     Switzerland |   .2521252   .3720001     0.68   0.498    -.4769816
>     .9812319
                         Belgium |   .3097169   .3003147     1.03   0.302     -.278889
>     .8983229
```

```
                 Czech Republic |   .1794089   .3796688     0.47   0.637    -.5647283
>     .9235461
                        Slovenia |   1.628394     .36302     4.49   0.000     .9168876
>       2.3399
------------------------------------+----------------------------------------------------
  ____________
                          /phi1_1 |          1  (constrained)
                          /phi1_2 |   .8351313   .0371097    22.50   0.000     .7623977
>     .9078649
                          /phi1_3 |   .6311646   .0294805    21.41   0.000     .5733838
>     .6889454
                          /phi1_4 |   .3539225   .0241565    14.65   0.000     .3065766
>     .4012683
                          /phi1_5 |          0  (base outcome)
------------------------------------+----------------------------------------------------
  ____________
                          /theta1 |    34.1627   12.56533     2.72   0.007     9.535102
>     58.79029
                          /theta2 |   30.13715   10.41483     2.89   0.004     9.724452
>     50.54985
                          /theta3 |   24.64583   7.889743     3.12   0.002     9.182215
>     40.10944
                          /theta4 |   14.65465   4.426552     3.31   0.001     5.978768
>     23.33053
                          /theta5 |          0  (base outcome)
------------------------------------+----------------------------------------------------
  ____________
(sphus=Poor is the base outcome)
results_imp1_ste_22_5.xls
dir : seeout

Predictive margins                                      Number of obs = 11,204
Model VCE: Robust

1._predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2._predict: Pr(sphus==Very good), predict(pr outcome(#2))
3._predict: Pr(sphus==Good), predict(pr outcome(#3))
4._predict: Pr(sphus==Fair), predict(pr outcome(#4))
5._predict: Pr(sphus==Poor), predict(pr outcome(#5))

------------------------------------+----------------------------------------------------
  ____
                                    |            Delta-method
                                    |     Margin   std. err.      z    P>|z|     [95% conf. inter
> val]
------------------------------------+----------------------------------------------------
  ____
      _predict#rhfo#ghto            |
            1#None#None             |   .0365705    .002669    13.70   0.000     .0313394    .041
> 8017
      1#None#At least 1             |   .0456516   .0047969     9.52   0.000     .0362498    .055
> 0534
      1#At least 1#None             |    .027625   .0028195     9.80   0.000     .0220989     .03
> 3151
1#At least 1#At least 1             |   .0330118   .0049995     6.60   0.000     .0232131    .042
> 8106
            2#None#None             |   .1152968   .0049132    23.47   0.000     .1056671    .124
> 9265
      2#None#At least 1             |   .1319555   .0079074    16.69   0.000     .1164572    .147
> 4538
      2#At least 1#None             |   .0967452   .0056217    17.21   0.000     .0857269    .107
> 7635
2#At least 1#At least 1             |   .1082092   .0095424    11.34   0.000     .0895064    .126
> 9121
            3#None#None             |   .3917333   .0079745    49.12   0.000     .3761035     .40
> 7363
      3#None#At least 1             |   .4105339   .0108107    37.97   0.000     .3893453    .431
> 7224
      3#At least 1#None             |    .365584   .0091041    40.16   0.000     .3477403    .383
> 4278
3#At least 1#At least 1             |   .3824273   .0139023    27.51   0.000     .3551793    .409
> 6753
```

```
            4#None#None |   .3414112   .0074138    46.05   0.000     .3268805    .355
> 9419
      4#None#At least 1 |   .3205101     .01095    29.27   0.000     .2990486    .341
> 9717
      4#At least 1#None |   .3618085   .0089776    40.30   0.000     .3442128    .379
> 4042
4#At least 1#At least 1 |   .3496281   .0122817    28.47   0.000     .3255563    .373
> 6998
            5#None#None |   .1149882   .0054243    21.20   0.000     .1043567    .125
> 6197
      5#None#At least 1 |   .0913489    .009235     9.89   0.000     .0732486    .109
> 4493
      5#At least 1#None |   .1482373   .0085845    17.27   0.000     .1314119    .165
> 0626
5#At least 1#At least 1 |   .1267236    .015549     8.15   0.000     .0962481    .157
> 1991
------------------------+----------------------------------------------------------
  ____
slog22_5.xls
dir : seeout
      name:  The_value_of_a_helping_hand_sup
       log:  C:\Users\hansg\Documents\working\SHARE data\weighting\output_sup_5.smcl
  log type:  smcl
 closed on:   5 Apr 2026, 14:55:18
------------------------------------------------------------------------------------
```